Fiscal Year:	FY 2013	**Task Last Updated:**	FY 02/05/2013
PI Name:	Baulch, Janet E. Ph.D.		
Project Title:	Space Radiation Effects on Genomic and Epigenetic Stability: Influence of Radiation Quality and Genetic Predisposition		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92697	**Congressional District:**	45
Comments:	NOTE: PI previously at University of Maryland; moved to UCA Irvine in summer 2013 (Ed., 11/11/13)		
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Space Radiobiology NNJ10ZSA001N
Start Date:	03/01/2011	**End Date:**	07/11/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/11/2013 due to PI move to UCA, Irvine (Ed., 11/11/13) NOTE: Correction to Period of Performance per C. Guidry/JSC (Ed., 10/4/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX11AC55G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Astronauts traveling in space are exposed to a variety of radiations. It is known that exposure to ionizing radiation can lead to an increased risk for developing cancer. Since there is currently no way to protect astronauts, it is important that the carcinogenic risk of exposures to these radiations is understood. The delayed, deleterious effects of radiation exposure are grouped under the terms radiation-induced genomic and epigenomic instability. These mechanisms are considered to be early steps in the carcinogenic process. Epigenetic mechanisms are thought to play a significant role in carcinogenesis by regulating gene expression and genome stability. These epigenetic mechanisms include DNA methylation, miRNA mediated gene silencing, and histone modification. There are also genetic factors in mouse and man that determine the relative sensitivity of their cells to radiation-induced effects and ultimately their cancer risk. This project will develop assays that will provide functional biomarkers for exposure that are also indicators of carcinogenic risk. We will accomplish these objectives by evaluating mice with differing sensitivities to radiation for epigenomic instability as measured by changes in miRNA and DNA methylation profiles following exposure to iron, silicon or oxygen ion radiations. The effect of age on the epigenetic radiation induced response will be evaluated using mice of relatively younger and older ages. [Editor's note 11/11/2013: See grant NNX13AK70G with the same title for subsequent reports, due to PI move to University of California, Irvine, in summer 2013]
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The goal of this grant is to evaluate the effect of space radiation exposure on epigenetic profiles with a focus on the role for genetic factors, age and radiation linear energy transfer (LET). Some of the research and technology questions that this proposal addresses include the following: 1) the probabilities for increased carcinogenic risk following relatively low dose radiation exposure as a function of NASA's operational parameters (radiation quality, age at exposure, genetic factors, tissue specific response); 2) the development of quantitative, functional biomarkers for carcinogenic risk; 3) a better understanding of the molecular mechanisms involved in cancer initiation and progression. Understanding the cancer risk associated with exposure to either terrestrial and space radiation is important for human health. Epigenetic abnormalities are thought to play a key role in the earliest steps of carcinogenic initiation. Epigenetic changes are mitotically and meiotically heritable, stable alteration in gene expression that include DNA methylation, histone modifications and RNA associated gene silencing. This project tests the hypothesis that radiation exposure can affect epigenetic profiles and increase carcinogenic risk. We further hypothesize that genetic factors and age at the time of exposure impact the epigenetic radiation response. Specific Aim 1 will evaluate radiation-induced alterations in miR profiles and gene expression. For each tissue, dose, and time point, genome-wide miR profiles will be analyzed. In parallel, global mRNA profiles will be analyzed to provide information regarding gene expression. Together these arrays will allow us to identify up-regulated miRNA that correlate with down-regulated target mRNA, or vis versa. Mining these data will allow us to identify affected pathways that are relevant to cancer and the radiation response. Specific aim 2 will evaluate the effect of irradiation on the induction of aberrant DNA methylation. Alterations in DNA methylation have emerged as one of the most consistent molecular markers in multiple cancers and for carcinogenic initiation and progression. We will measure genome wide methylation by evaluating repeat element methylation. These repeat elements make up a significant portion of the mammalian genome and aberrant repeat element methylation has been linked to genomic instability, altered gene expression and cancer. We will also evaluate promoter methylation for genes associated with increased carcinogenic risk and the radiation response. Using these three strategies to evaluate tissue specific DNA methylation in irradiated mice we will be able to survey genome wide changes that have been linked to carcinogenic processes. As a whole, this project uses a systems biology approach that will provide information regarding tissue-specific effects of radiation LET, genetic factors, age, and post-irradiation time on miRNA-mediated gene regulation as it applies to the radiation response and radiation-induced carcinogenesis. This benefits not only our astronauts, but those of us on Earth, by increasing our understanding of the molecular mechanisms of cancer and the radiation response.
Task Progress:	The first aim of this project is to evaluate radiation-induced alterations in miR profiles and gene expression. For each tissue, dose, and time point, genome-wide miR and mRNA profiles will be analyzed. Together these arrays will allow us to identify changes in gene expression that are effected by changes in miR. Mining these data will allow us to identify affected pathways that are relevant to cancer and the radiation response. The second aim of this project is to evaluate the effect of irradiation on the induction of altered DNA methylation. Alterations in DNA methylation have emerged as one of the most consistent molecular markers in multiple cancers and for carcinogenic initiation and progression. During 2012 we made significant progress towards achieving these goals particularly aim 2. 1. Participation in the NSRL 12A and 12C runs. During these runs we completed exposures of younger and older C3H/HeN and C57Bl/6 male mice to Fe ions (600 MeV/n, LET 175 keV/um) and Si ions (350 MeV/n, LET 65 keV/um) using doses of 10, 30 and 100 cGy. Tissues have been harvested from 1, 7, 30, and 120 days post-irradiation and archived for molecular analyses (miR and mRNA arrays and analysis of DNA methylation). 2. Completion of development and validation of pyrosequencing assays for a panel of genes involved in the carcinogenic process and the radiation response (including MGMT, p16 and DAPK1). 3. Completion of development and validation of pyrosequencing assays for two repeat elements as a measure of genome-wide DNA methylation. The A1 monomer is a long interspersed nucleotide element (LINE) sub-family with at least 9,400 copies in the mouse genome. The TF monomer is another LINE sub-family with approximately 550 copies in the mouse genome. 4. Ongoing development and validation of high resolution melt curve (HRM) methodology to evaluate levels of DNA methylation. 5. Initiation of collaboration with Dr. Rithidech to evaluate DNA methylation in hematopoietic stem cells exposed to heavy ions. 6. Analysis of DNA methylation for liver and lung tissues from our first cohort of mice, 10-12 week old C3H/HeN male mice exposed to 600 MeV/n Fe ions. These data demonstrate clear tissue-specific, locus-specific, time-dependent low

	dose effects on DNA methylation profiles. Upon completion of the qRT-PCR analyses of gene expression for affected loci, this work will be written up for publication. 7. Completion of a study evaluating the roles of mitochondrial dysfunction and epigenetic regulation in radiation induced genomic instability. Thomas, S, Yang, A, Morgan, WF, Baulch, JE. Quantitative Proteomic Analysis of Mitochondrial Proteins Reveals Pro-Survival Mechanisms in the Perpetuation of Radiation Induced Genomic Instability. Free Radical Biology and Medicine. 2012 August;53:618-628. [Editor's note 11/11/2013: See grant NNX13AK70G with the same title for subsequent reports, due to PI move to University of California, Irvine, in summer 2013]
Bibliography Type:	Description: (Last Updated: 05/01/2017)
Articles in Peer-reviewed Journals	Thomas SN, Waters KM, Morgan WF, Yang AJ, Baulch JE. "Quantitative proteomic analysis of mitochondrial proteins reveals prosurvival mechanisms in the perpetuation of radiation-induced genomic instability." Free Radical Biology and Medicine. 2012 Aug 1;53(3):618-28. Epub 2012 Apr 19. http://dx.doi.org/10.1016/j.freeradbiomed.2012.03.025 ; PubMed PMID: 22569412 , Aug-2012